

EX 9 : FORWARD CHAINING

Exercise 1: Simple Inference

`:- dynamic known/1.`

`% Initial knowledge`

`start :- assert(known(a)), assert(known(b)), infer.`

`% Rule`

`infer :- known(a), known(b), \+ known(c), assert(known(c)), write(''Inferred
c'), nl.`

`% Query`

`% ?- start.`

Output:

`39 ?- consult(['c:/Users/HDCO422166/Downloads/simple.pl']).
true.`

`40 ?- start.`

Inferred c
`true.`

Exercise 2: Fire Alert System

`:- dynamic known/1.`

`% Rule to infer fire`

`check :- known(smoke), known(heat), assert(known(fire)), write(''Fire
Alert!'), nl.`

`% Start rule`

`start :- assert(known(smoke)), assert(known(heat)), check.`

`% Query`

`% ?- start.`

Output:

`45 ?- consult(['c:/Users/HDCO422166/Downloads/fire.pl']).
true.`

```
46 ?- start.  
Fire Alert!  
true.
```

Exercise 3: Water Purity Detection

```
:- dynamic known/1.
```

```
% Rule to infer pollution
```

```
infer_pollution :- known(cloudy_water), known(bad_smell), assert(known(polluted)),  
write('&#39;Water is polluted&#39;), nl.
```

```
start :- assert(known(cloudy_water)), assert(known(bad_smell)), infer_pollution.
```

```
% Query
```

```
% ?- start.
```

```
Output:
```

```
49 ?- consult(['c:/Users/HDCO422166/Downloads/water.pl']).  
true.
```

```
50 ?- start.
```

```
Water is polluted  
true.
```